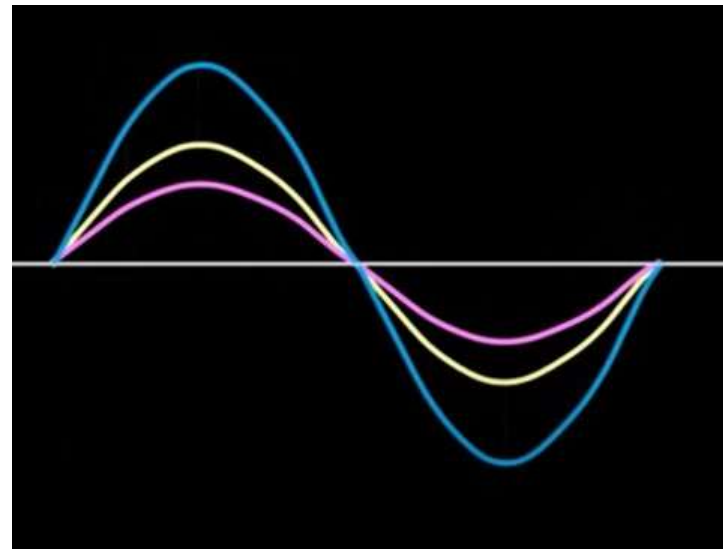
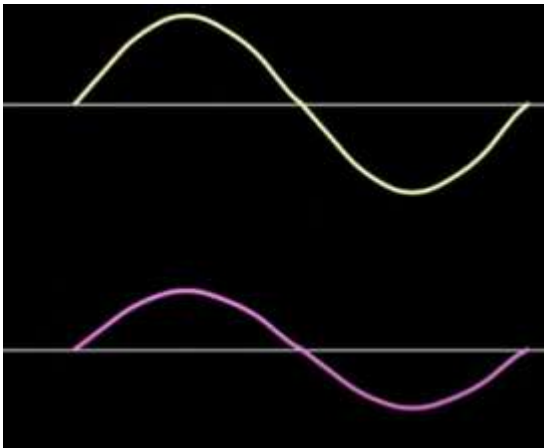


phase velocity definition :

Phase velocity is defined as a rate at which the phase of a wave propagates in space . It is analogous to the rate of propagation of a dot along the wave itself.

Two waves
with same wavelength!

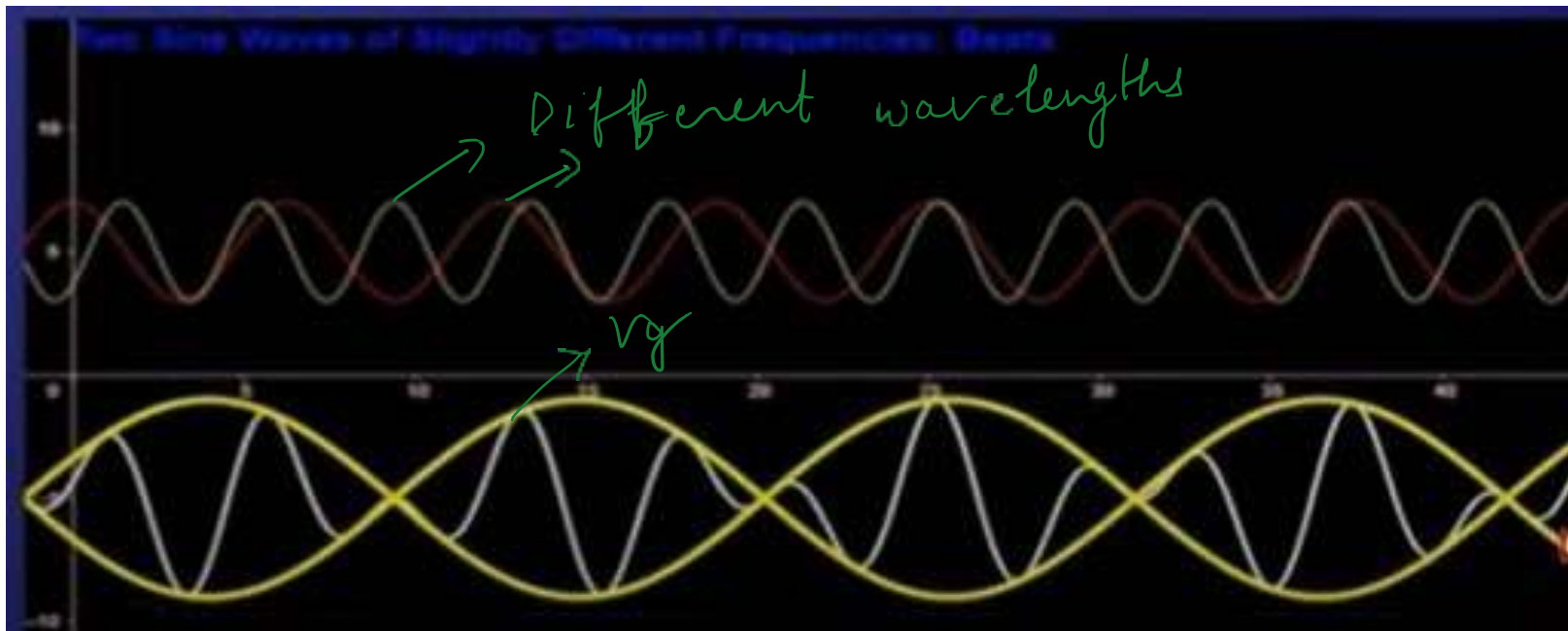
So same velocity v_p



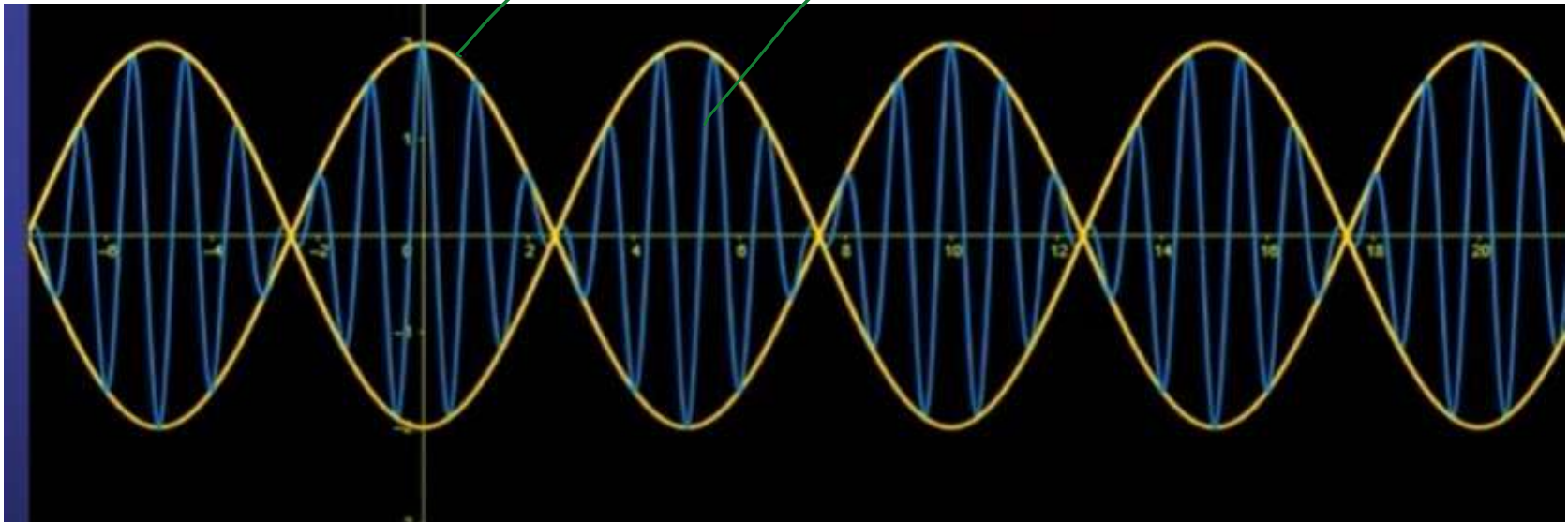
group velocity definition :

The **group velocity** of a wave is the velocity with which the overall envelope shape of the wave's amplitudes also known as the modulation or envelope of the wave—propagates through space.

A wave group consists of a superposition of waves with different wavelengths, with the amplitude and phase of each component wave adjusted so that the waves interfere constructively over a small region of space. Outside of this region the combination of waves produces a net amplitude that approaches zero rapidly as a result of destructive interference.

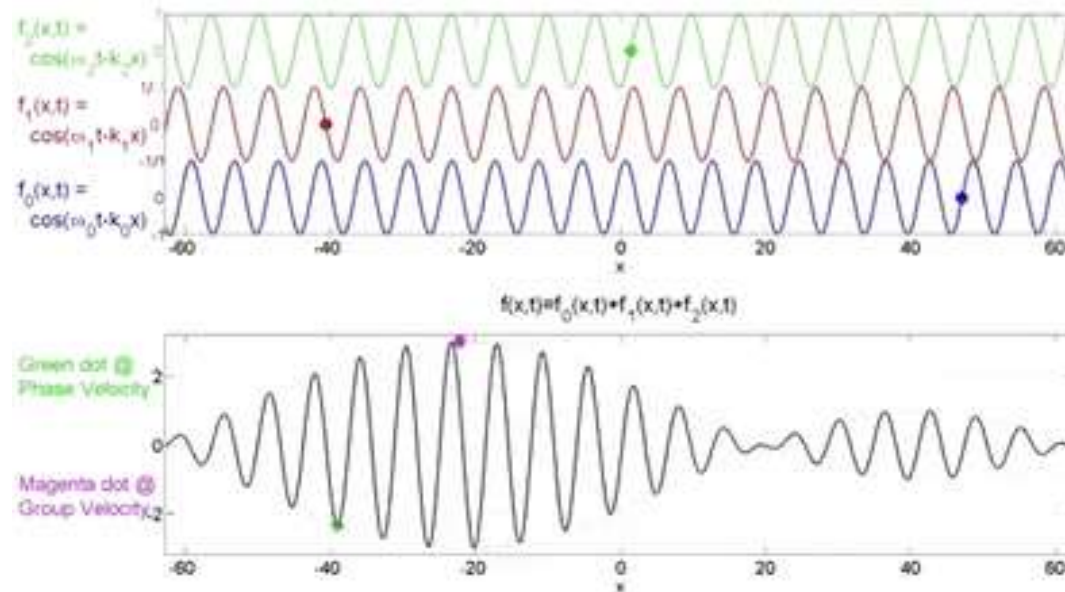


$$v_p = \omega / \beta$$



$$v_g = \frac{d\omega}{d\beta}$$

The Group Velocity



This is the velocity at which the overall shape of the wave's amplitudes, or the wave 'envelope', propagates. (= *signal velocity*)

$$\boxed{v_p = \omega / \beta}$$

$$\left[v_g = \frac{d\omega}{d\beta} \right]$$

$$v_g = \frac{d}{d\beta} (v_p \beta)$$

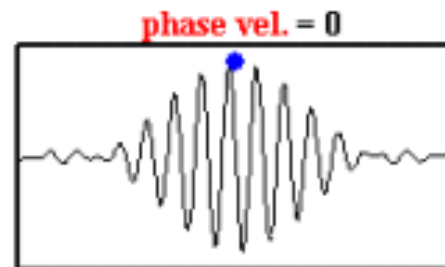
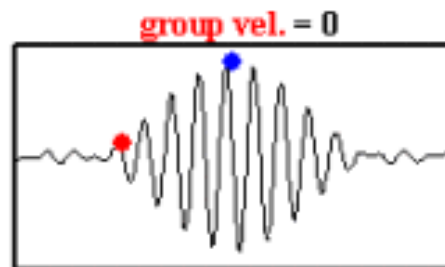
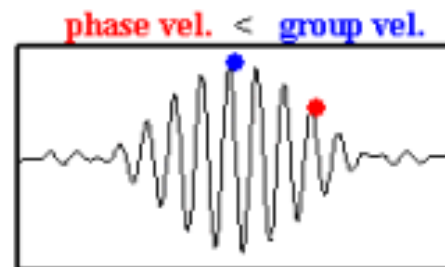
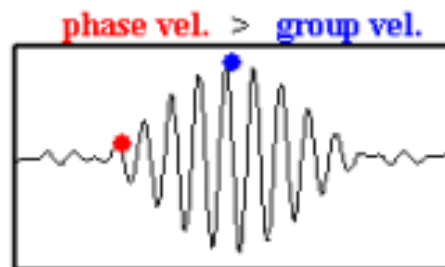
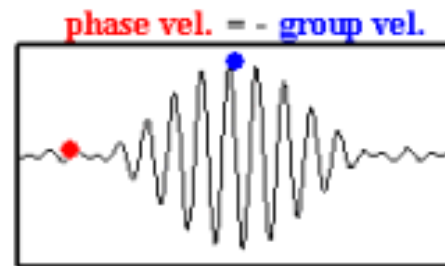
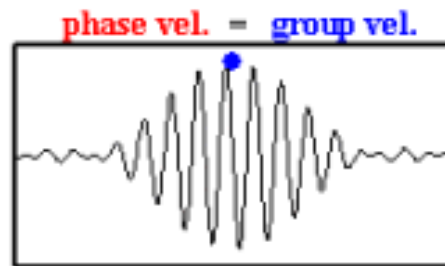
$$\left[v_g = v_p + \beta \frac{dv_p}{d\beta} \right]$$

$$v_g = v_p + \frac{2\pi}{\lambda} \frac{dv_p}{-\frac{2\pi}{\lambda^2} d\lambda}$$

$$\boxed{v_g = v_p - \lambda \frac{dv_p}{d\lambda}}$$

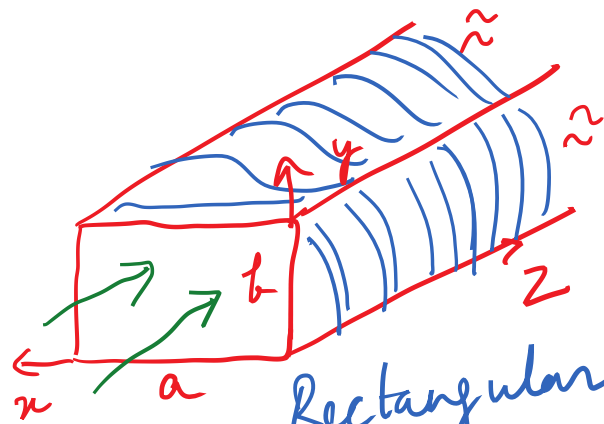
Relation ship b/w
phase velocity &
group velocity

Different Cases for V_p and V_g



Rectangular waveguide resonators

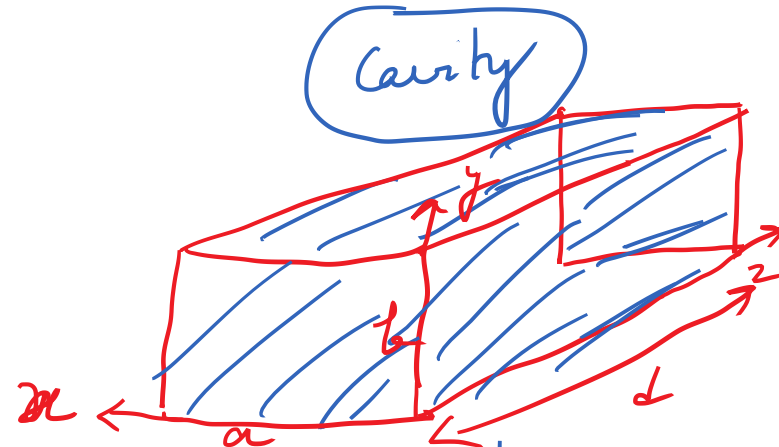
Rectangular cavity



$$[f > f_c]$$

Rectangular waveguide

TM mode was done



FW

$$E_{z_{fwd}} = C \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta'z}$$

BW

$$E_{z_{bwd}} = D \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{j\beta'z}$$

$$\underline{H_z = 0}$$



Analyzing boundary condⁿ at $z=0$ or $z=d$

$$E_{\text{fwd}} = E_{\text{bwd}} \quad \text{so } \angle = \Delta$$

Total E_z field +

$$E_{z_{\text{tot}}} = c' \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \left[e^{-j\beta'z} + e^{j\beta'z} \right]$$

$$E_{z_{\text{tot}}} = 2c' \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \cos(\beta'z)$$

to max^m
at $z=d$ $\beta'd = l\pi$

$$E_{z_{\text{tot}}} = 2c' \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \cos\left[\frac{l\pi z}{d}\right] \quad \text{so } \beta' = \frac{l\pi}{d} \quad \left[\beta' = \frac{l\pi}{d} \right]$$

In case of rectangular waveguide!

$$\omega_{\text{cutoff}} = \frac{1}{\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

$$f_{\text{cutoff}} = \frac{c}{2} \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}}$$

In cavity resonator!

$$\omega^2 \mu\epsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{l\pi}{d}\right)^2$$

$$\omega_{\text{reso}} = \frac{1}{\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{l\pi}{d}\right)^2}$$

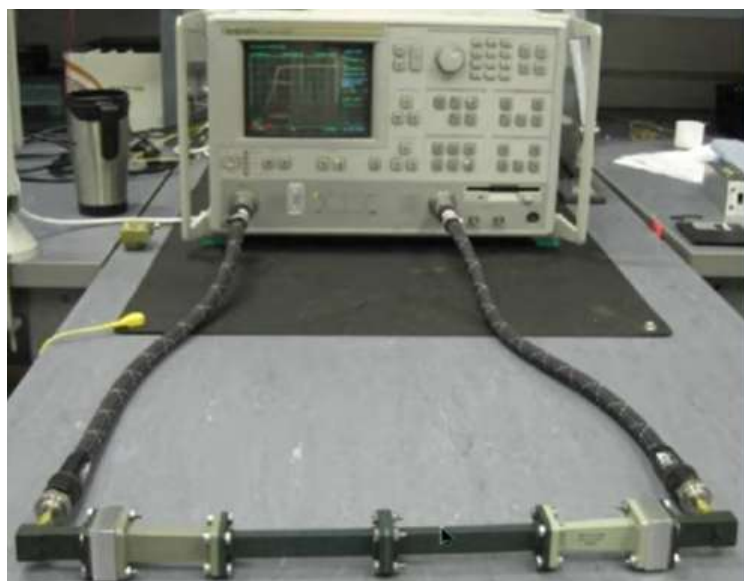
$$\left[f_{\text{reso}} = \frac{c}{2} \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2} + \frac{l^2}{d^2}} \right]$$

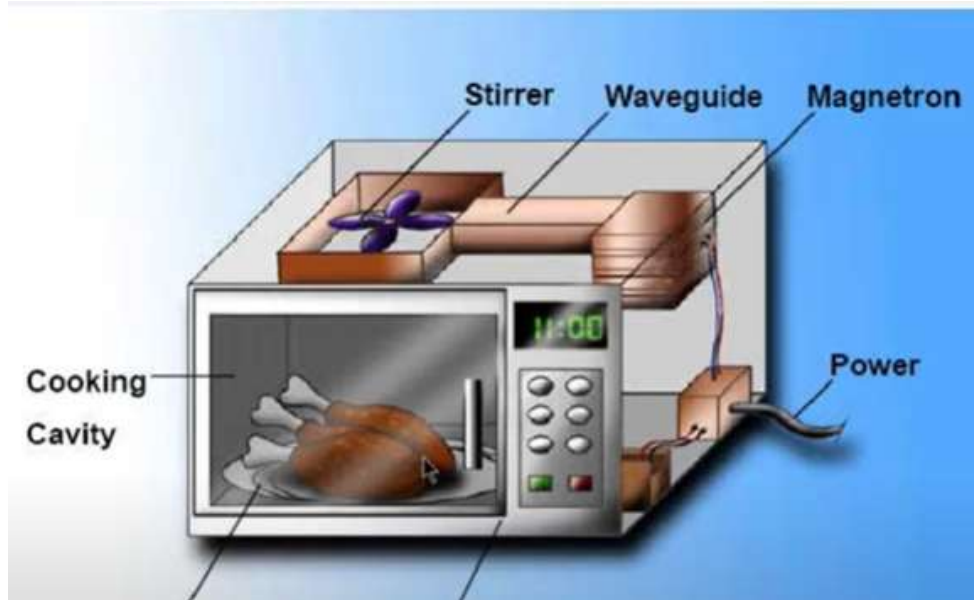
TE_{mn} ,

TM_{mn} -

TM_{mnl} -

TE_{mnl} ✓





Applications of Cavity Resonator

- ❑ **Oscillators:** Provide stable frequencies for signals.
- ❑ **Filters:** Used to select specific frequencies from a broad spectrum.
- ❑ **Microwave Communication:** Essential components in microwave transmitters and receivers.
- ❑ **Spectroscopy:** Used in devices like masers and lasers for precise frequency control.

